MSA UNIVERSITY
جامعة أكتوبر للعلوم الحديثة والآداب

Software Documentation document for project primary tumors

Ziad Gamaleldin,

Ziad Hassan,

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Supervised By: Dr .Moamen Zaher

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Introduction

This dataset is about primary tumors. So that, the purpose of this project is to give the helping hand to the people who suffer from primary tumors and they are not aware about the presence of it. However, this data helps in the early diagnoses of the tumors before it migrate in to other locations in the human body and become life threatening.

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1. Problem Definition

- Primary tumor are locations in body where the tumor first appeared and from there they start to metastasize to other parts of the body, so we chose to work on this health care domain because about 90% of the cases die due to late diagnoses. However, we are motivated to do this project because the primary tumor is a serious problem that can treated when diagnosed faster before it could metastasize from one organ to another and lead to the death of the person, so we are providing some information that would help in the detection of these tumors faster.

2. Dataset Information

- The data was collected from orange data mining, and it consists of 6 features and 339 instances each, these features points out to data about patients who suffered from primary tumors, the main goal is the early diagnoses of these tumors based on 6 relevant features which are:

    1) Histologic type
    2) Degree of life
    3) Sex
    4) Axillar
    5) Mediastinum
    6) Neck

primary	histologic_type	degree_of_diffe	sex	axillar	mediastinum	neck
corpus uteri	adeno	well	female	no	no	no
ovary	adeno	well	female	no	no	no
ovary	adeno	well	female	no	no	no
ovary	adeno	well	female	no	no	no
ovary	adeno	well	female	no	no	no
ovary	adeno	well	female	no	no	no
ovary	adeno	well	female	no	no	no
ovary	adeno	well	female	no	no	no
ovary	adeno	well	female	no	no	no
prostate	adeno	well	male	no	no	no
prostate	adeno	well	male	no	yes	no
prostate	adeno	well	male	no	no	no
prostate	adeno	well	male	no	yes	yes
kidney	adeno	well	male	no	no	no
kidney	adeno	well	female	no	no	no
kidney	adeno	well	male	yes	no	no
kidney	adeno	well	male	no	no	no
pancreas	adeno	well	female	no	no	no
pancreas	adeno	well	male	no	no	no
pancreas	adeno	well	female	no	no	no
pancreas	adeno	well	male	no	no	no
rectum	adeno	well	female	no	no	no
rectum	adeno	well	male	no	no	no
rectum	adeno	well	male	no	no	no
colon	adeno	well	male	no	no	no

Figure 1. Data set information

3. Methodology

- First, we got our data from orange data mining. After that we used the ranking widget to rank and filter our data from irrelevant features. Then, we used select column widget in order to remove the unnecessary columns.

- After that, we modeled the primary tumor dataset by using several algorithms as the logistic regression, SVM, neural network, and CN2 rule induction as well as comparing between these algorithms, then we tested them by classifying the data into 95% training and 100 testing.

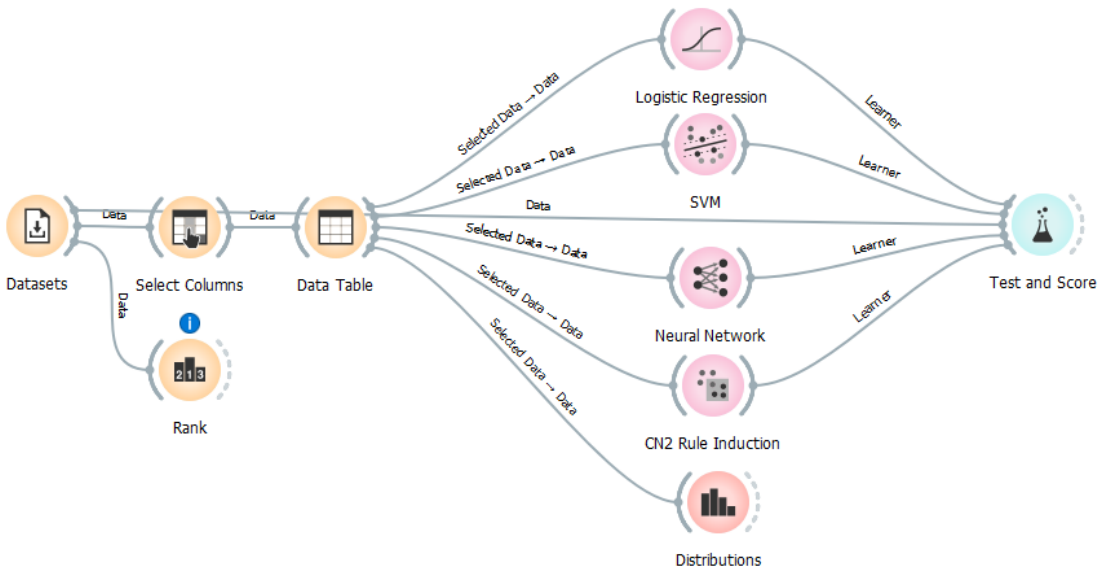

Figure 2. Processes

4. Results

- In the results of the graph, it's concluded that the primary tumors are likely to develop more at the lungs followed by the stomach and the ovary and pancreas as it's shown in the distribution visualization.

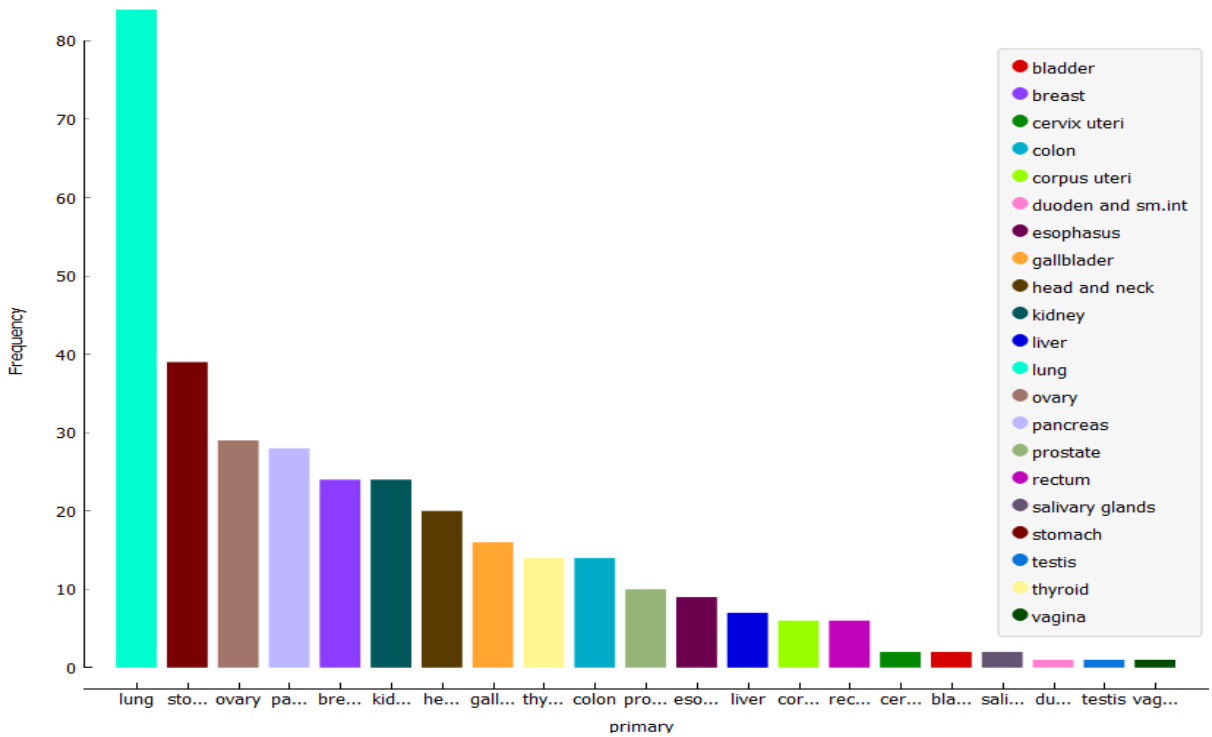

Figure 3. Distribution graph

- After comparing between the algorithms, Its concluded that the neural network is the most accurate algorithm by achieving 0.993 followed by the CN2 rule inducer which achieved 0.992 followed by SVM which achieved 0.973 followed by Logistic regression which achieved the least accurate result which is 0.942.

Evaluation Results

Model	AUC	CA	F1	Precision	Recall
SVM	0.973	0.844	0.839	0.858	0.844
Neural Network	0.993	0.870	0.867	0.875	0.870
Logistic Regression	0.942	0.614	0.573	0.563	0.614
CN2 rule inducer	0.992	0.867	0.871	0.895	0.867

Figure 4. Evaluation Results

5. Team Members

Name	ID	Week	Lecture group	Lab group
Ziad Gamaleldin	210309	Week one	C2	Group B
Ziad Hassan	210663	Week one	C2	Group B

6. References

- https://www.scirp.org/(S(351jmbntvnsjt1aadkposzje))/reference/ReferencesPapers.aspx?ReferenceID=1838945
- https://www.researchgate.net/publication/335314617_Primary_Tumor_Culture_and_Method_Development_Studies_of_Patients_with_Lung_Cancer